GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.2

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LS 1.1.2

Introduction to Pharmacokinetics & Pharmacodynamics

OBJECTIVES

At the end of the Lecture, students shall be able:

- To explain the terms Pharmacokinetics and Pharmacodynamics easily
- To discuss the importance of pharmacokinetics and pharmacodynamics in the determination of drug dosage regimens correctly
- To outline the four pharmacokinetic processes of Absorption, Distribution, Metabolism and Excretion correctly
- To differentiate between drug actions and drug effects easily

OBJECTIVES

- To outline the mechanisms by which drugs produce their effects
- To describe the relationship between pharmacokinetics and pharmacodynamics
- To describe the time course of drug action
- To describe the relationship between drug dose and clinical response
- To define the terms Therapeutic Range and Therapeutic Index
- Calculate the Therapeutic Index

Pharmacokinetics & Pharmacodynamics

INTRODUCTION:

Pharmacokinetics refers to the movement of the drug

1. into the body
2. through the body
3. out of the body

Taking into account the processes of drug input (absorption), drug distribution and elimination of the drug from the body (metabolism and excretion)

Pharmacodynamics describes the effects of drugs and the mechanisms by which drugs exert their effects

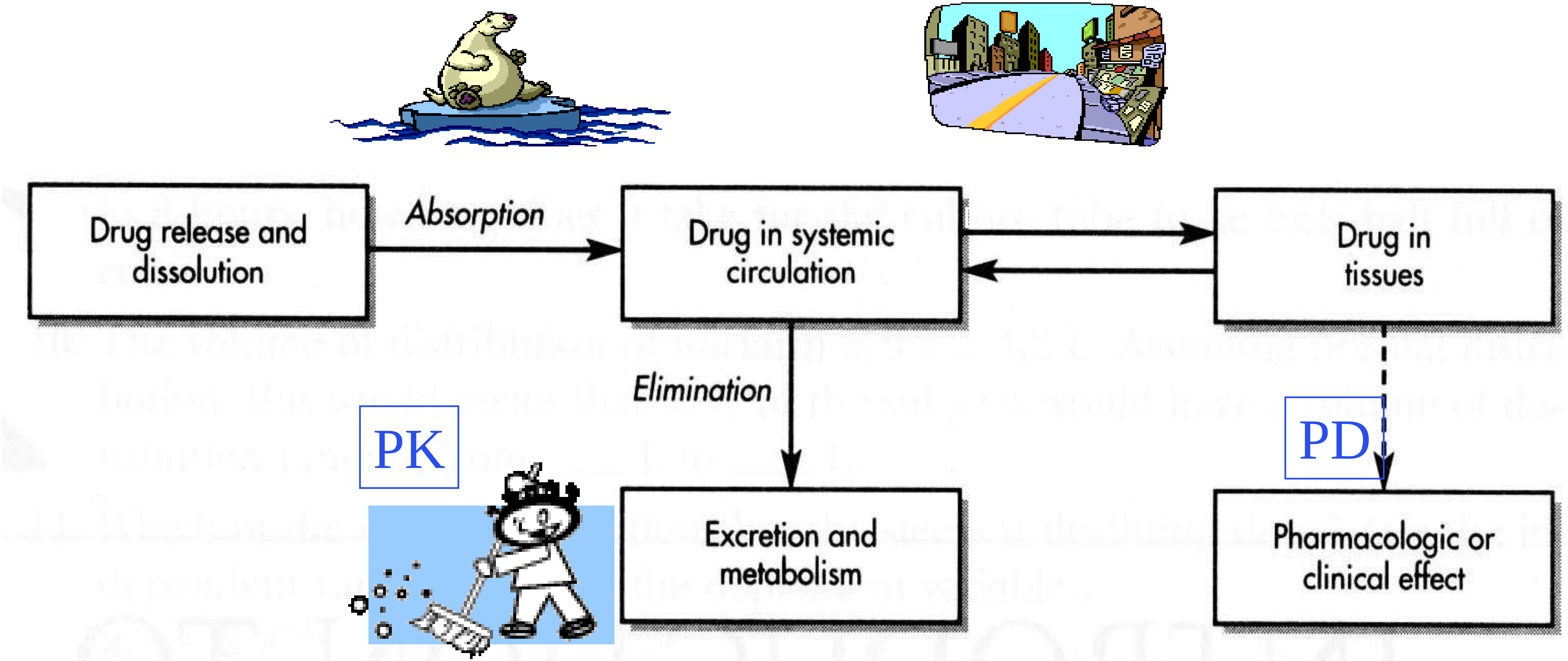


Figure 2-1. Scheme demonstrating the dynamic relationship between the drug, the drug product, and the pharmacologic effect.

What happens to the drug in the body?

Pharmacokinetics

Pharmacokinetics is the study of the processes involved in the disposition of drugs, namely absorption, distribution, metabolism and excretion (ADME)

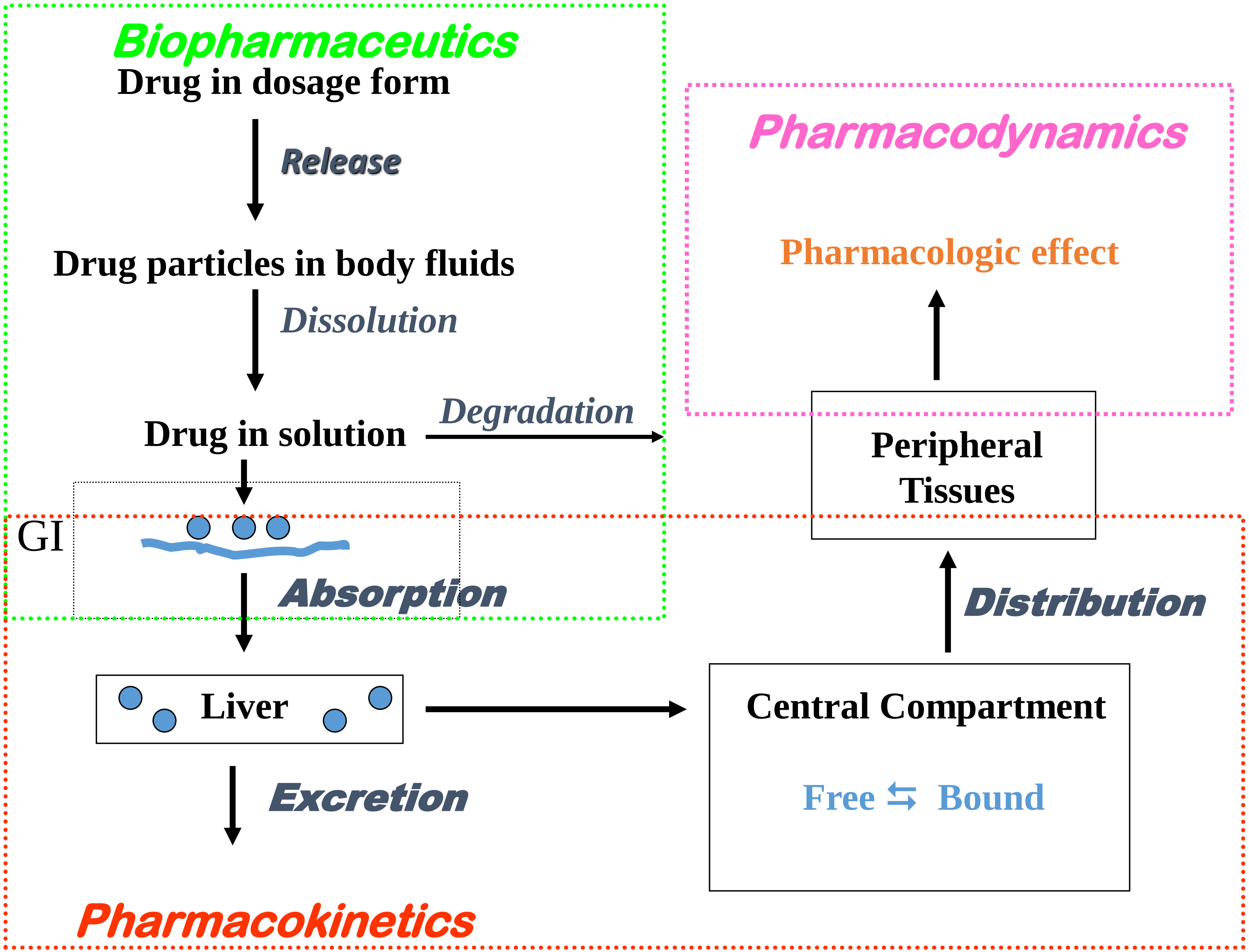
This includes:

- ❖ The rate and extent to which drugs are absorbed into the body and distributed to the body tissues
- ❖ The rate and pathways by which drugs are eliminated from the body (metabolism and excretion)
- ❖ The relationship between time and plasma drug concentration

PHARMACOKINETICS

Understanding the **pharmacokinetic** processes is important because:

- It forms the basis on which the optimal dose regimen is chosen
- It explains the majority of the inter-individual variation in the response to drug therapy



Biopharmaceutics
Drug in dosage form
Release
Drug particles in body fluids
Dissolution
Drug in solution
Degradation
GI
Absorption
Liver
Excretion
Pharmacodynamics
Pharmacologic effect
Peripheral Tissues
Distribution
Central Compartment
Free ⇆ Bound
Pharmacokinetics

PHARMACOKINETICS: ADME

Absorption

This is the process by which drugs enter the body

- When given by any route other than intravenously, drug molecules must cross tissue membranes (e.g. skin epithelium, subcutaneous tissue, gut endothelium, capillary wall) to enter the blood

Distribution

This is the process by which drugs move around the body

- After entering the blood, drug molecules must cross capillary walls to enter the tissues, reach cell membranes and enter cells

PHARMACOKINETICS: ADME …. CONT'D

Metabolism

The process by which drugs are chemically altered to make them sufficiently water-soluble for excretion in urine or feaces (via the biliary tract)

- Metabolism occurs in a variety of body organs and tissues, but chiefly in the liver, gut wall, kidney and skin

PHARMACOKINETICS: ADME …. CONT'D

Excretion

The process by which drugs leave the body

- Drugs that are sufficiently water-soluble will be excreted unchanged in the urine
- Lipid-soluble drugs must be modified to water-soluble metabolites before excretion via the kidney or into the intestine via the bile

PHARMACOKINETICS: APPLICATION

- For a drug to produce a beneficial effect without intolerable unwanted effects, it needs to be present in the right place, at the right concentration and for the right duration
- Changes in drug tissue and plasma concentration over time is determined by the pharmacokinetic profile of the drug
- Knowledge of drug pharmacokinetics is therefore required for rational dosing; dose regimens are determined from drug pharmacokinetic parameters

PHARMACODYNAMICS

Pharmacodynamics is the study of the biochemical and physiological effects of drugs and their mechanisms of action

- Involves the study of the actions and effects of drugs
- Involves the study of the interactions between drugs and cellular macromolecular components which result in biological effects

DRUG ACTION VERSUS DRUG EFFECT

Drug action

Mechanisms by which the drug produces a response in living organisms

Drug effect

The observable consequence of a drug action

Example: The action of penicillin is to interfere with cell wall synthesis in bacteria and the effect is the death of the bacteria

MECHANISM OF ACTION VERSUS MODE OF ACTION

Mechanism of action: Refers to the specific biochemical interaction through which a drug produces its pharmacological effect (= Drug Action)

Mode of action: Describes a functional or anatomical change, resulting from the exposure of a living organism to a drug (= Drug Effect)

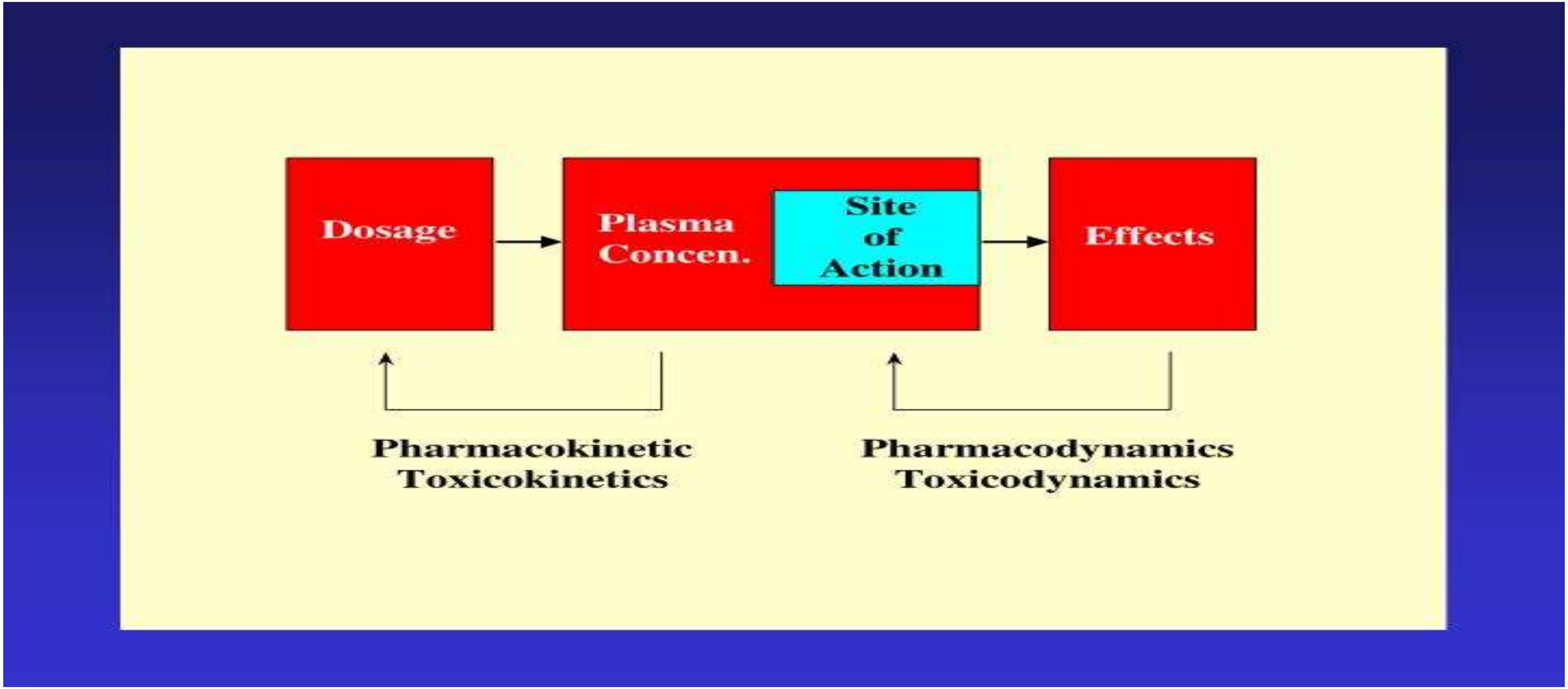
Mechanism of action describe changes at molecular level while mode of action describes changes at cellular level

MECHANISMS OF ACTION

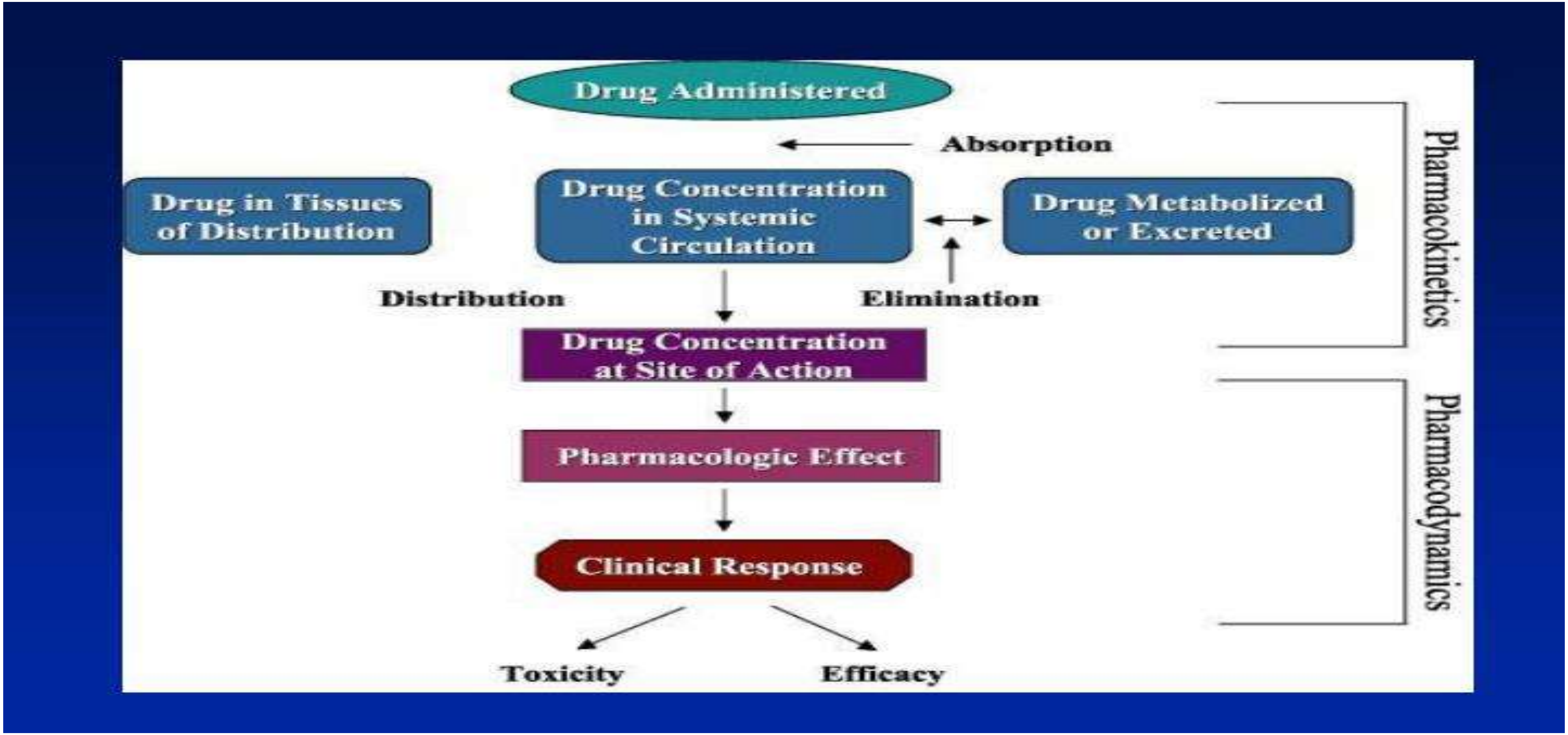
Most drugs act by interacting with macromolecular targets (mostly proteins) which include receptors, enzymes, carriers, membrane bound transport proteins and pumps, and ion channels

- Drugs action may lead to stimulation or depression of normal physiological functions. Stimulation increases the rate of activity while depression reduces the rate of activity.
- Drugs may act by increasing or decreasing enzymes, hormones or other chemical mediators of biological activities

RELATIONSHIP BETWEEN DRUG PHARMACOKINETICS AND DRUG PHARMACODYNAMICS



RELATIONSHIP BETWEEN DRUG PHARMACOKINETICS AND DRUG PHARMACODYNAMICS …. CONT'D



RELATIONSHIP BETWEEN DRUG PHARMACOKINETICS AND DRUG PHARMACODYNAMICS …. CONT'D

- The pharmacokinetic properties of a drug determine the concentration of drug achieved in plasma and tissues, including the site of action of the drug
- Drug effects are proportional to the concentration of drug at the action site
- An equilibrium is established between plasma drug concentration and concentration at the active site, therefore there is a correlation between drug effects and plasma drug concentration

RELATIONSHIP BETWEEN PHARMACOKINETICS AND PHARMACODYNAMICS

- Pharmacokinetics determines the change in plasma drug concentration with time, and therefore how rapidly and for how long the drug will appear at the target organ
- Pharmacodynamics determines the magnitude of drug effect in relation to plasma drug concentration
- Plasma drug concentration forms the link between drug dosing and effect

RELATIONSHIP BETWEEN PHARMACOKINETICS AND PHARMACODYNAMICS .... CONT'D

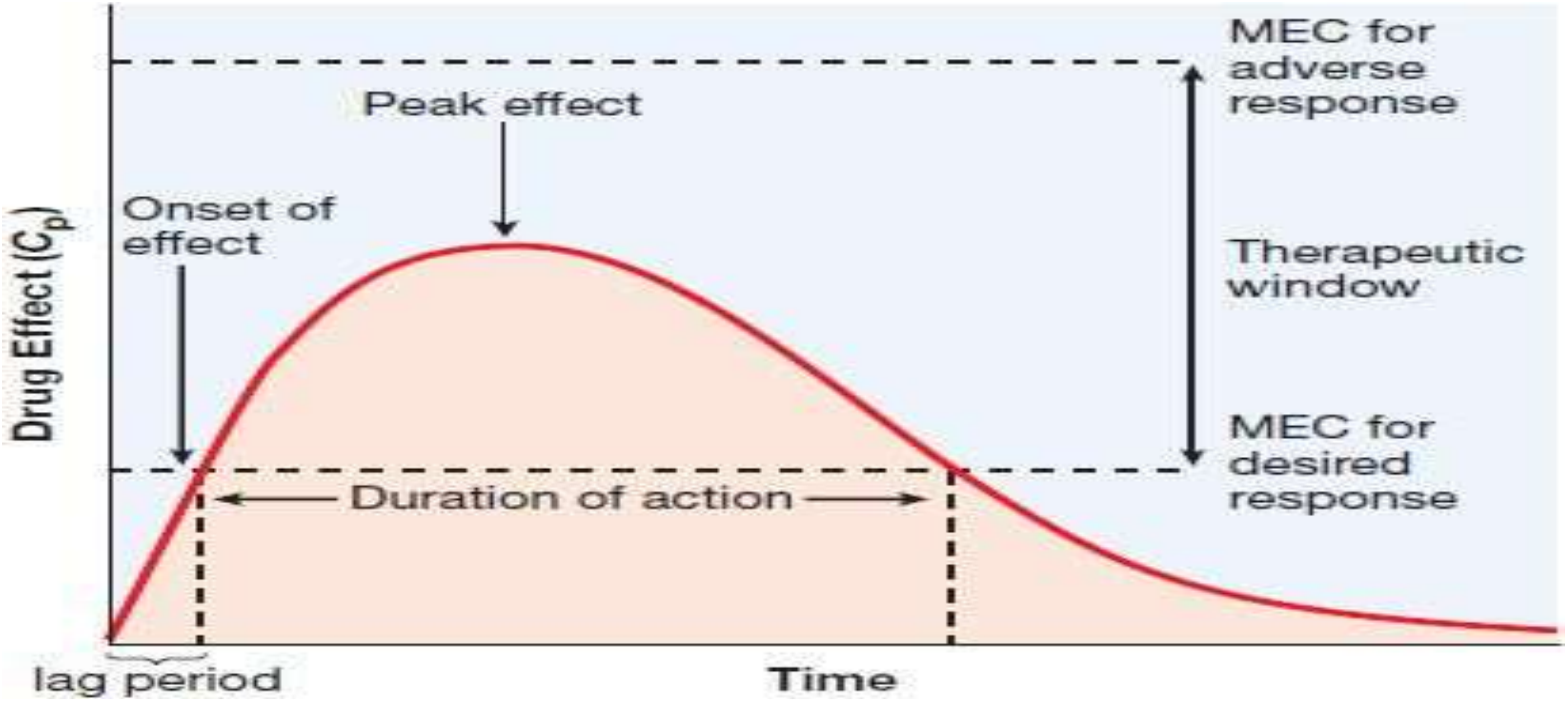
- The relationship between dose and effect can be separated into two components:

    1. Pharmacokinetic (dose-concentration)

    2. Pharmacodynamic (concentration-effect)

- Plasma drug concentration provides the link between pharmacokinetics and pharmacodynamics and is therefore the focus when determining a dose regimen

TIME COURSE OF DRUG ACTION



TIME COURSE OF DRUG ACTION …. CONT’D

Onset of drug action

The time it takes after the drug is administered to reach a concentration that produces a response

Duration of action

The time during which the drug is present in a concentration large enough to produce a response

Peak effect

The time it takes for the drug to reach its highest effective action

TIME COURSE OF DRUG ACTION …. CONT'D

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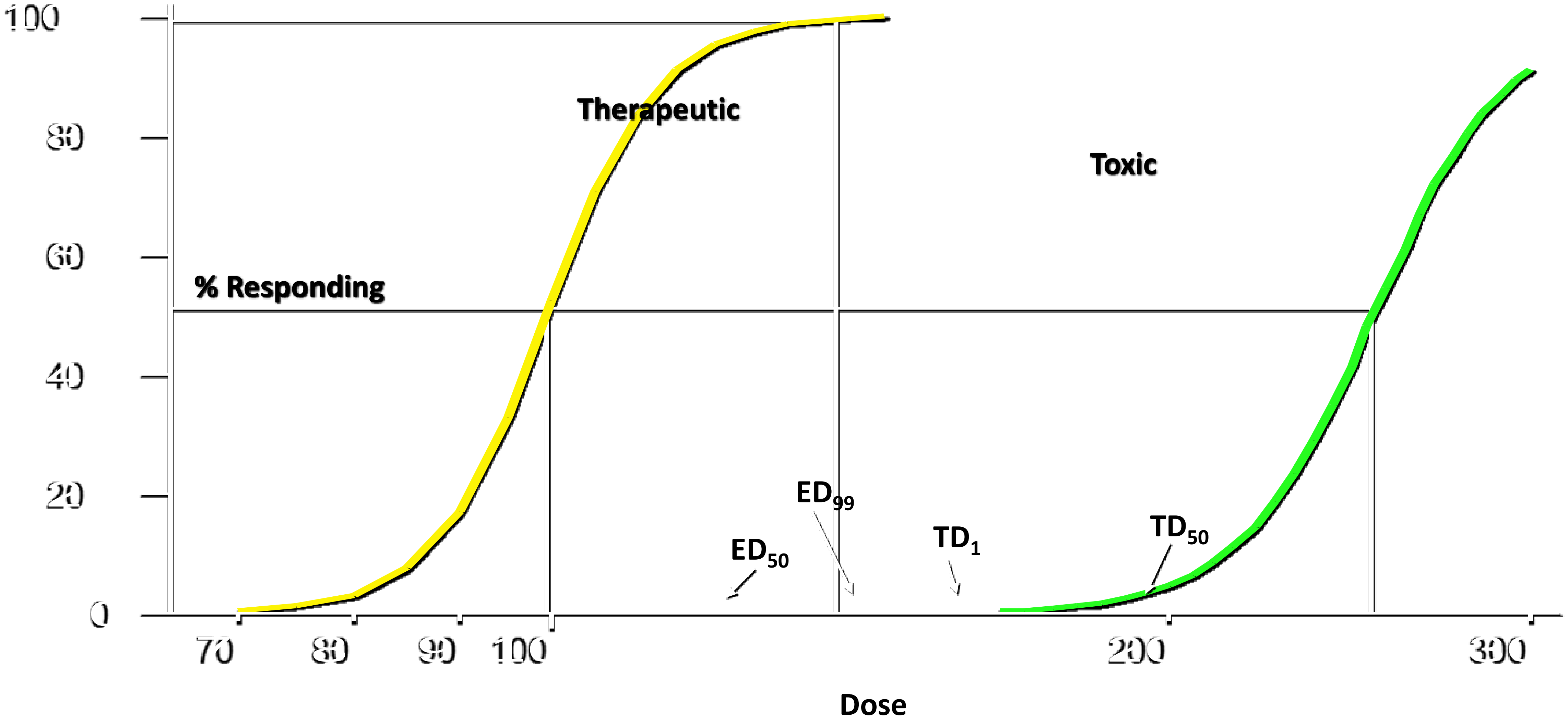
RELATIONSHIP BETWEEN DRUG DOSE & CLINICAL RESPONSE

Minimum effective dose (MED): The lowest dose level of a drug that provides a clinically significant response

- There is a threshold for each drug to produce a response
- Doses of drugs below that threshold will produce no observable effect

Maximum tolerated dose (MTD): The highest dose of a drug that will produce the desired effect without unacceptable toxicity

RELATION BETWEEN DRUG DOSE & CLINICAL RESPONSE …. CONT'D



RELATIONSHIP BETWEEN DRUG DOSE & CLINICAL RESPONSE …. CONT'D

- **Median effective dose (ED_{50}):** The dose of a drug that produces a specified effect in 50% of the population that takes that dose
- **Median toxic dose (TD_{50}):** The dose of a drug that produces a particular toxic effect in 50% of the population that takes that dose
- **Median lethal dose (LD_{50}):** The dose of a drug required to kill 50% of a given test population

THERAPEUTIC RANGE (WINDOW) AND THERAPEUTIC INDEX

Therapeutic range (therapeutic window)

- The concentration of drug between minimum effective concentration and maximum safe concentration is known as therapeutic range.
- Range of the dose levels from the MED to the MTD

Therapeutic index

- MTD/MED
- TD_{50}/ED_{50}

THERAPEUTIC RANGE (WINDOW) AND THERAPEUTIC INDEX …. CONT'D

- The relative safety of a drug can be expressed by its therapeutic index
- A drug with a wide therapeutic index has a high safety margin and is relatively safe; the lethal dose is much larger than the therapeutic dose
- A drug with a narrow therapeutic index is more dangerous for the patient because small increases over normal doses may induce toxic reactions

LEARNING OUTCOMES

Students must be able to comprehend:

- The basic concepts of pharmacokinetics and pharmacodynamics
- The relationship between pharmacokinetics and pharmacodynamics
- Therapeutic applications of pharmacokinetics and pharmacodynamics
- Time course of drug action
- Relationship between drug dose and clinical response
- The concept of therapeutic index

QUIZ

1. Pharmacokinetics is:

a) The study of biological and therapeutic effects of drugs b) The study of absorption, distribution, metabolism and excretion of drugs c) The study of mechanisms of drug action d) The study of methods of new drug development

2. What does “pharmacokinetics” include?

a) Complications of drug therapy b) Drug biotransformation in the organism c) Influence of drugs on metabolism processes d) Influence of drugs on genes

3. What does “pharmacokinetics” include?

a) Pharmacological effects of drugs b) Unwanted effects of drugs c) Chemical structure of a medicinal agent d) Distribution of drugs in the organism

4. What does “pharmacokinetics” include?

a) Localization of drug action b) Mechanisms of drug action c) Excretion of substances d) Interaction of substances